

# SPO Source Library — Visual Operation Guide

SCREEN-BY-SCREEN ARCHITECTURAL MAPPING & INGESTION RULES

## 🎯 THE MAIN JUICE (CORE SYSTEM PHILOSOPHY)

The intentional engineering objective of this interface is to orchestrate **NotebookLM background automation** to yield structured JSON summaries. The researcher operates at a high strategic level: they are **not interested in reading raw JSON outputs**, nor do they spend time manually editing text records here.

The pure tactical value of this page is tracking the pipeline status: **Which reference theses have been processed, what is currently running, how is the batch progress moving, and exactly what failed.**

## 1. Screenshot-Driven Interface Walkthrough

Below is a precise structural breakdown of the screen states using the system's execution snapshots to map how files transition into structured knowledge blocks.

### SNAPSHOT 1: Baseline Layout [image\_e01904.png]

State: Fresh Initialization

Component: Master Accordions

The application starting framework. Both **Card 01 (Drive Setup & Indexing)** and **Card 02 (Source Library)** exist in an optimized, completely minimized state. This ensures that when a researcher launches the application, they are greeted by a clean workspace without horizontal or vertical clutter before paths are established.

## SNAPSHOT 2: Discovered Local Trackers [image\_e333fe.jpg & image\_e07703.jpg]

State: Directory Discovery

Metrics: Ingestion Ready (0 / 13 Indexed)

Once a local parent folder path (e.g., .../Shodhganga\_Downloads) is designated and scanned, Card 01 transitions into a wide management table. The screen exposes the structural fleet of 13 separate works detected across the filesystem.

At this stage, the status column reads a grey Idle, and every row mounts a bright blue Run action indicator alongside a global Run All Unindexed master command button. This signals to the researcher that the local mapping is confirmed, but no external NotebookLM pipeline context has been created yet.

## SNAPSHOT 3: Component Delta Matrix [image\_e01926.jpg & image\_e0108b.jpg]

State: Surgical Sub-File Selection

Context: Active Checkbox Arrays

Expanding an unindexed or processed thesis row exposes its internal structure. An academic thesis is not a single file block; it consists of multiple nested PDF files (e.g., 01 title.pdf, chap 1.pdf, works cited.pdf).

As captured in **image\_e0108b.jpg**, the interface empowers the user to surgically select which specific component files are bundled and uploaded. The system visualizes these files through a rigid, multi-color state matrix:

### ■ Green Highlight

**Already in notebook — will be skipped.** Items that match existing file signatures within NotebookLM's container, preventing redundant network consumption and double-uploads.

### ■ White Highlight

**Selected — new, will be uploaded.** Fresh files detected in the folder sequence that will be included in the upcoming processing run.

### ■ Muted Grey

**Excluded by you.** Components that have been unselected by the user to prevent unwanted content from polluting the analytical model.



#### SNAPSHOT 4: Hydrated Knowledge Base [image\_e00208.jpg]

State: Parsed JSON Serialization

Component: Card 02 Content Hub

When the background worker completes its automation sequence, Card 02 dynamically unpacks. It displays the literal entries generated inside the underlying backend JSON. It strips away raw file names and replaces them with clean structural metadata (Author profiles, full dissertation titles, and chronologically mapped chapters).

**Imperfect Parsing Resiliency Rule:** If due to a background extraction bug or user configuration oversight, an unwanted component PDF was grouped and sent to NotebookLM, Card 02 is architected to display that item faithfully as it exists in the JSON output. The layout deliberately **\*\*omits an inline delete switch\*\***. To maximize file synchronization, the user simply corrects the checkbox allocations in Card 01 and clicks Re-run to execute an atomic clean overwrite.

## 2. Deep-Dive UX Operational Layout Rules

The interface relies on discrete interactive workflows to bridge Card 01 configuration with Card 02 storage knowledge:



### Clickable Title Links

Immediately following a successful pipeline sequence, the thesis titles inside Card 01 transform from plain static headers into active external hyperlinks. Clicking these anchors securely redirects the researcher's browser tab straight into the respective live notebook workspace inside the **NotebookLM cloud interface** for independent verification.



### Viewport Auto-Scroll Arrow

Positioned directly to the right of the row-level Re-run button in Card 01 is an indicator arrow icon. Clicking this icon performs an instant, zero-friction viewport jump, scrolling the page down to the exact coordinates where that specific thesis's parsed structural metadata cards reside within the massive container of Card 02.



### Pipeline Status Telemetry

The core telemetry engine monitors system processes across three dedicated variables:

- **Active Processing:** Real-time progress fractions (e.g., 1 / 13 INDEXED) pulsing bright green/blue as automated browsers cycle uploads.
- **Failures & Stalls:** Absolute error flags indicating exact network, file, or token exceptions at the specific row level.
- **Sync Overwrites:** Re-triggering a run instantly safely wipes the old flat-file cache for that thesis ID, maintaining strict crash safety.